

YANYING WEI

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EDUCATION

University of Chinese Academy of Sciences (Top Graduate Schools in China)

Beijing, China

M.Sc. in Finance (Applied Economics | Natural Resource Management & Industrial Economics)

September 2019 - June 2022

Core courses:

GPA: 3.8/4

- Advanced Macroeconomics
- Advanced econometrics
- Environmental Economics
- Optimization & Simulation
- Advanced Microeconomics
- Financial Engineering
- Business Data Analytics
- Big Data Mgmt & Apps

Guangxi University (World-Class Universities, Project 211)

Nanning, China

B.Sc. in Finance (Applied Economics | Financial Mathematics)

September 2015 - June 2019

Core courses:

GPA: 3.3/5

- Mathematical Analysis
- Operations Research
- Stochastic Processes
- Database
- Advanced Algebra
- Behavioral Finance
- Risk Management
- Statistics

PUBLICATIONS

- Liu, J., **Wei, Y.**, Wang, S., & Dong, Z. (2026). *Decision Rights, Fiscal Flows, and County Fiscal Expenditure: A Systems Perspective on China's Province-Managing-County Reform*. *Systems*, 14(7), 819. doi: 10.3390/systems14070819.
- Wang, T., & **Wei, Y.** (2026). *The Impact of the Low-Carbon City Pilot Policy on Energy Intensity: Evidence from a Staggered Difference-in-Differences Design*. *Land*, 15(6), 913. doi: 10.3390/land15060913.
- Liu, J., **Wei, Y.**, Wang, S., & Dong, Z. (2026). *Fiscal Intermediaries, Transfer Delivery, and Sustainable Local Growth: Evidence from China's Province-Managed-County Reform*. *Sustainability*, 18(11), 5276. doi: 10.3390/su18115276.
- **Wei, Y.**, Shi, L., Jiang, Y., & He, Q. (Peer Review). *Analyzing the Impact of Electricity Reforms on Energy Supply Security: Green finance as a Policy Perspective*. *Transportation Research Interdisciplinary Perspectives*. JCR: Q1.
- **Wei, Y.** (2026). *Optimizing regional carbon peak pathways under uncertainty: An integrated Tapio-LMDI-robust optimization framework with evidence from Guangxi, China*. *Energy Conversion and Management: X*, 31, 102058. doi: 10.1016/j.ecmx.2026.102058. JCR: Q1.
- Shi, L., & **Wei, Y.** (2025). *Dynamic Network-LSTM Model for Quantitative Analysis of Infrastructure REITs' Impact on Urban Rail Transit Debt Ratios*. In *Proceedings of the 2025 2nd International Conference on Economic Data Analytics and Artificial Intelligence (EDAII 2025)*. ACM. doi: 10.1145/3789297.3789298.
- Dong, Z., **Wei, Y.**, Ren, C., & Wang, S. (2022). *Comprehensive land improvement innovation for rural revitalization: Public value creation and realization*. *Resources Science*, 44(7), 1305–1315. doi: 10.18402/resci.2022.07.01.

RESEARCH EXPERIENCE

Research on Carbon Emissions in Guangxi Province Based on LMDI and Tapio Decoupling Models

January 2024 - Current

- Reviewed and synthesized over 100 relevant academic papers to inform methodology and contextualize findings.
- Collected and processed data from over 50 governmental and industrial sources, ensuring high data accuracy and reliability.
- Conducted in-depth analysis using LMDI and Tapio decoupling models to evaluate the carbon intensity of Guangxi's economy.
- Identified key drivers of carbon emissions and quantified their impact, showing a 15% reduction in carbon intensity over five years.

Efficiency Measurement in Guangxi's Science and Technology Finance: Based on DEA Model

October 2023 - June 2024

- Led a study on efficiency measurement in Guangxi's science and technology finance using DEA modeling, encompassing literature review, data collection, model analysis, and conclusion.
- Utilized the DEA model to assess the efficiency of science and technology finance in Guangxi, identified inefficiencies, and quantified their impact, offering actionable insights for policymakers and stakeholders.
- Mentored one student in producing graduation theses based on project findings, resulting in recognition for outstanding theses.

Research on Coupling Coordination of Land Use, Socio-Economic, and Ecological Systems

January 2021 - June 2022

- Conducted a comprehensive literature review using R language with the Bibliometrix package and CiteSpace, analyzing over 500 academic sources.
- Utilized Principal Component Analysis (PCA) to filter influential factors, and expanded the study's scope to include a supply chain perspective in assessing system coupling coordination.
- Quantified the coupling coordination degree of land use, socio-economic, and ecological systems, providing a measurable indicator of system integration and sustainability, with a demonstrated improvement of 20% in system efficiency.

RESEARCH FUNDINGS

- **Principal Investigator.** *Value-Capture Challenges and Incremental Hub Reconfiguration Pathways for the Guangxi Corridor Economy under the New International Land-Sea Trade Corridor.* Supported by the Guangxi Zhuang Autonomous Region Philosophy and Social Sciences Office (Grant No. 25JYF186), October 2025 - October 2028.
- **Principal Investigator.** *Feixiangjia: A Solid Waste Management Platform Tailored for SMEs.* Supported by Nanning University (Grant No. S2024-1154-9017X), June 2024 - December 2027.
- **Principal Investigator.** *Research on the Decoupling State and Driving Mechanism of Carbon Emissions in Guangxi under the Dual-Carbon Target.* Supported by the Department of Education of Guangxi Zhuang Autonomous Region (Grant No. 2024-KY-1871), January 2024 - December 2025.
- **Principal Investigator.** *Digital Empowerment for Bank Loans and Regulatory Innovation.* Supported by Nanning University (Grant No. S2023-1154-9126), January 2024 - December 2024.
- **Approved Project.** *Python and AI Programming Smart Course Initiative.* Supported by Nanning University, 2025.

PROFESSIONAL EXPERIENCE

Nanning University, School of Digital Economy

Nanning, China

Lecturer of Fintech

September 2022 - Current

- **Core Course Instruction:** Delivered lectures in Python Programming & AI, Python Web Scraping & Data Collection Lab, Financial Big Data Technologies & Applications, Big Data Applications in Finance, and Econometrics to over 200 students annually, integrating advanced theory with practical applications.
- **Curriculum Development:** Enhanced course materials with cutting-edge financial technology and Python programming, maintaining relevance to the digital economy and boosting student engagement and comprehension.
- **Research and Mentorship:** Initiated and completed 5 of 8 proposed projects at the Digital Economy Research Institute, published findings, and provided academic and career guidance to over 50 students, contributing to a 95% thesis defense pass rate.

Shanxi Land Institute of Natural Resources and Green Development

Beijing, China

Staff of Technology R&D

June 2021 - July 2022

- **Project Leadership:** Led significant projects, such as the *Study on Comprehensive Land Improvement in Huangtong Town, Lingao County, Hainan Province*, and the *Construction of a Town of Incense in Huangtong Town*, securing a 3 billion yuan investment.
- **Policy Development:** Spearheaded integrated land policy projects in Xinjiang, Guangxi, and Yunnan, enhancing land utilization efficiency by an estimated 10%.
- **Publication Contribution:** Authored a module in the *Natural Resources Lecture Series*, published by China Earth Publishing House, on fundamental information science principles.

HONORS & AWARDS

- **Nanning University Outstanding Graduation Thesis:** Outstanding Advisor Award, June 2025 (Title: "Research on Price Prediction of CSI 300 Index Based on K-Means and LSTM Models", Student: Jinmei Huang).
- **Nanning University Outstanding Graduation Thesis:** Outstanding Advisor Award, June 2024 (Title: "Research on the Measurement and Current Situation of Technology Finance Efficiency in Guangxi", Student: Yinyin Liao).
- **ICBC Cup National College Students Financial Technology Innovation Competition:** Outstanding Instructor Award, December 2022, 2023 & 2025 (Highly recognized competition in China).
- **Nanning University:** General Typical Case of Research-Informed Teaching, 2025.
- **Institute of Scientific and Technical Information of China:** Top Academic Paper in China's Excellent Scientific and Technical Journals, 2023.
- **Nanning University Teaching Skills Competition:** Third Prize, July 2022.
- **National Undergraduate Mathematical Modeling Competition:** Second Prize of Guangxi Region, July 2019.
- **American Mathematical Contest in Modeling:** "S" Prize, January 2017.

SKILLS & PROFESSIONAL CERTIFICATE

- **Languages:** Mandarin (Native), English (Fluent, GRE 321).
- **Developing Skills:** C, Python, Matlab, SQL, R, SPSS, Stata, LaTeX, Git.
- **Frameworks/Technologies:** Scrapy, Pandas, Selenium, Request.